Unit 23: Antiderivatives — Full Solutions

Warm-up

1. $\int x^3\,dx$

Bump the exponent up by 1 (to 4), then divide by that new exponent:

$$\int x^3\,dx = \frac{x^4}{4} + C$$

2. $\int 5\,dx$

A constant antidifferentiates to that constant times x:

$$\int 5\,dx = 5x + C$$

3. $\int x^4\,dx$

$$\int x^4\,dx = \frac{x^5}{5} + C$$

4. $\int 3x^2\,dx$

The constant 3 just tags along:

$$\int 3x^2\,dx = 3 \cdot \frac{x^3}{3} + C = x^3 + C$$

5. $\int (x^2 + 2x)\,dx$

Antidifferentiate each term separately:

$$\int (x^2 + 2x)\,dx = \frac{x^3}{3} + x^2 + C$$

6. $\int (4x^3 - 3x^2 + 1)\,dx$

$$\int (4x^3 - 3x^2 + 1)\,dx = x^4 - x^3 + x + C$$

7. $\int \sqrt{x}\,dx$

Rewrite as a power first: $\sqrt{x} = x^{1/2}$.

$$\int x^{1/2}\,dx = \frac{x^{3/2}}{3/2} + C = \frac{2}{3}x^{3/2} + C$$

Standard

8. $\int \sin x\,dx$

$$\int \sin x\,dx = -\cos x + C$$

9. $\int (3\cos x - 2\sin x)\,dx$

$$\int (3\cos x - 2\sin x)\,dx = 3\sin x - 2(-\cos x) + C = 3\sin x + 2\cos x + C$$

10. $\int \sec^2 x\,dx$

$$\int \sec^2 x\,dx = \tan x + C$$

11. $\int \frac{2}{x^3}\,dx$

Rewrite: $\frac{2}{x^3} = 2x^{-3}$.

$$\int 2x^{-3}\,dx = 2\cdot\frac{x^{-2}}{-2} + C = -x^{-2} + C = -\frac{1}{x^2} + C$$

12. $\int \left(x^2 - \frac{1}{x^2}\right)dx$

Rewrite: $x^2 - x^{-2}$.

$$\int x^2\,dx = \frac{x^3}{3}$$

$$\int -x^{-2}\,dx = -\frac{x^{-1}}{-1} + C = x^{-1} + C = \frac{1}{x} + C$$

Combining:

$$\int \left(x^2 - \frac{1}{x^2}\right)dx = \frac{x^3}{3} + \frac{1}{x} + C$$

13. $\int (\sec x\tan x + \csc x\cot x)\,dx$

$$\int (\sec x \tan x + \csc x \cot x)\, dx = \sec x + (-\csc x) + C = \sec x - \csc x + C$$

14. $\int (4x^3 - 6\sqrt{x})\, dx$

Rewrite $\sqrt{x} = x^{1/2}$.

$$\int 4x^3\, dx = x^4$$

$$\int -6x^{1/2}\, dx = -6 \cdot \frac{x^{3/2}}{3/2} + C = -4x^{3/2} + C$$

Combining:

$$\int (4x^3 - 6\sqrt{x})\, dx = x^4 - 4x^{3/2} + C$$

Challenge

15. $f'(x) = 3x^2 - 4x + 1,\ f(0) = 5.$

First, find the general antiderivative:

$$f(x) = x^3 - 2x^2 + x + C$$

Use $f(0) = 5$: $f(0) = 0 - 0 + 0 + C = C$. So $C = 5$.

$f(x) = x^3 - 2x^2 + x + 5.$

16. $f'(x) = 6x^2 - 2,\ f(1) = 3.$

General antiderivative:

$$f(x) = 2x^3 - 2x + C$$

Use $f(1) = 3$: $f(1) = 2 - 2 + C = C$. So $C = 3$.

$f(x) = 2x^3 - 2x + 3.$

17. $a(t) = 6t - 4,\ v(0) = 3,\ s(0) = 2.$

First antidifferentiate $a(t)$ to get the general velocity function:

$$v(t) = 3t^2 - 4t + C_1$$

Use $v(0) = 3$: $v(0) = 0 - 0 + C_1 = C_1$. So $C_1 = 3$.

$$v(t) = 3t^2 - 4t + 3$$

Now antidifferentiate $v(t)$ to get the general position function:

$$s(t) = t^3 - 2t^2 + 3t + C_2$$

Use $s(0) = 2$: $s(0) = 0 - 0 + 0 + C_2 = C_2$. So $C_2 = 2$.

$v(t) = 3t^2 - 4t + 3$ and $s(t) = t^3 - 2t^2 + 3t + 2$.

18. $f'(x) = 2\sin x + 3\cos x$, $f(0) = 4$.

General antiderivative:

$$f(x) = -2\cos x + 3\sin x + C$$

Use $f(0) = 4$: $f(0) = -2\cos(0) + 3\sin(0) + C = -2(1) + 3(0) + C = -2 + C$. So $-2 + C = 4 \Rightarrow C = 6$.

$f(x) = -2\cos x + 3\sin x + 6$.

19. $\displaystyle\int (x+1)(x-2)\,dx$

Expand first, since the power rule can't be applied directly to a product:

$$(x+1)(x-2) = x^2 - 2x + x - 2 = x^2 - x - 2$$

Now antidifferentiate term by term:

$$\int (x^2 - x - 2)\,dx = \frac{x^3}{3} - \frac{x^2}{2} - 2x + C$$

Applied

20. $a(t) = -32$, $v(0) = 48$, $s(0) = 0$.

Antidifferentiate acceleration to get velocity:

$$v(t) = -32t + C_1$$

Use $v(0) = 48$: $C_1 = 48$.

$$v(t) = -32t + 48$$

Antidifferentiate velocity to get position:

$$s(t) = -16t^2 + 48t + C_2$$

Use $s(0)=0$: $C_2=0$.

$$s(t)=-16t^2+48t$$

Find the maximum height: this occurs when $v(t)=0$.

$$-32t+48=0 \quad \Rightarrow \quad t=\frac{48}{32}=1.5 \text{ seconds}$$

$$s(1.5)=-16(2.25)+48(1.5)=-36+72=36$$

Answer: $v(t)=-32t+48$, $s(t)=-16t^2+48t$, and the maximum height is 36 feet, reached at $t=1.5$ seconds.

21. $MC(x)=3x^2-12x+15$, $C(0)=200$.

General antiderivative:

$$C(x)=x^3-6x^2+15x+K$$

(using K here to avoid confusion with the cost function's own name C)

Use $C(0)=200$: $K=200$.

$C(x)=x^3-6x^2+15x+200$.

22. $MR(x)=50-2x$, $R(0)=0$.

General antiderivative:

$$R(x)=50x-x^2+C$$

Use $R(0)=0$: $C=0$.

$R(x)=50x-x^2$.

23. $a(t)=-10$, $v(0)=60$.

General antiderivative:

$$v(t)=-10t+C$$

Use $v(0)=60$: $C=60$.

$$v(t)=-10t+60$$

The car stops when $v(t)=0$:

$$-10t+60=0 \quad \Rightarrow \quad t=6$$

Answer: $v(t) = -10t + 60$, and the car comes to a complete stop after 6 seconds.